Chapter 9 · Semester 1

DEFORMATION OF SOLIDS

Notes by Cikgu Suhaili

2 worked examples 1 simulation Exercises with answers

Deformation

Deformation

is any **change in the shape or size of an object** caused by an external force.

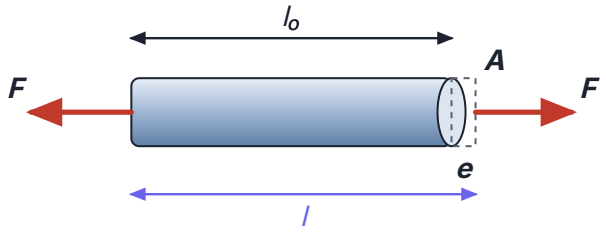
Elastic deformation

The object **regains its original shape and size** when the force is removed. It occurs when the material is deformed **within its elastic limit**.

Plastic deformation

The object **does not regain its original shape** when the force is removed. It occurs when the material is deformed **beyond its elastic limit**.

9.1 Stress and strain



A wire of original length l_o and cross-sectional area A, stretched by a force F perpendicular to A, extends by e.

Tensile stress, σ

is the **force per unit cross-sectional area** acting on a body that tends to deform it.

$$\sigma = \frac{F}{A}$$

- Scalar quantity.
- Dimension $\mathrm{M\,L^{-1}\,T^{-2}}$; unit N m^{-2} = Pa (kg m^{-1} s^{-2}).

Tensile strain, ε

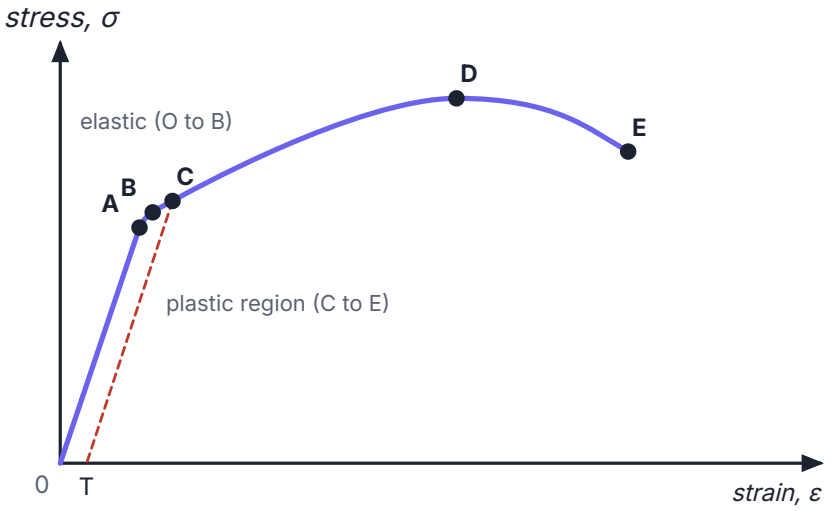
is the **ratio of the extension to the original length**.

$$\varepsilon = \frac{e}{l_0} = \frac{l - l_0}{l_0}$$

- Scalar quantity, no unit (dimensionless).
- l_0 = original length, l = final length, e = extension.

9.2 Force–extension and stress–strain graphs

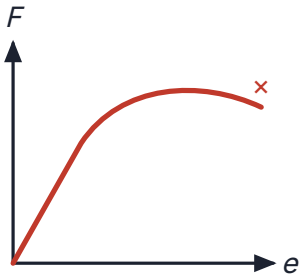
Stress–strain graph for a ductile metal



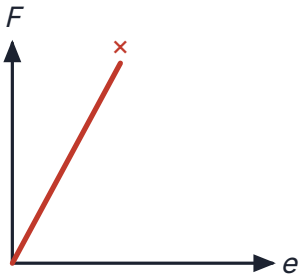
Point or region	Meaning
OA, A = **limit of proportionality**	Stress increases linearly with strain up to A. The straight line OA obeys **Hooke's law**: the restoring force is directly proportional to the extension.
B = **elastic limit**	Beyond B the material is permanently stretched and never regains its original length. In the region OB the material undergoes **elastic deformation**.
C = **yield point**	A small increase in stress produces a large increase in strain. When the stress is removed the material keeps a permanent strain (OT). Planes of atoms slide across each other and the wire thins.
CD = **plastic deformation**	As the stress increases, the strain increases rapidly.
D = maximum stress	The **breaking stress**, also called the **ultimate tensile strength** (UTS).
E = fracture point	The material breaks.

Unloading from a point beyond the elastic limit follows a line parallel to OA, so the material is left with a permanent strain OT.

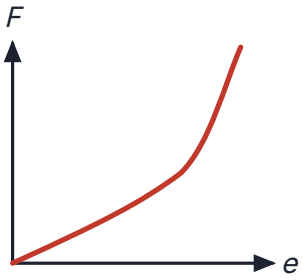
Force–extension graphs for ductile, brittle and polymeric materials



Ductile



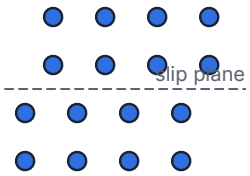
Brittle



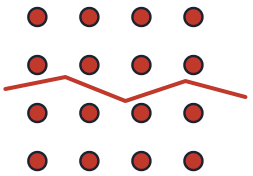
Polymer (rubber)

Material	Examples	Shape of the F–e graph
Ductile	copper, mild steel, aluminium	straight line, then a long plastic region before fracture
Brittle	glass, cast iron, ceramics	straight line almost up to fracture; snaps suddenly with almost no plastic extension
Polymeric	rubber	non-linear, S-shaped; very large extension, much of it recoverable

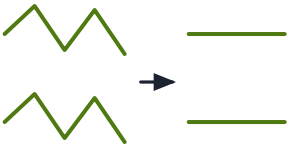
The gradient of the straight part is the stiffness. A steeper line (e.g. brittle glass) means a larger Young's modulus, but the material still fractures suddenly with little or no plastic deformation.



Ductile: planes slip



Brittle: crack runs through



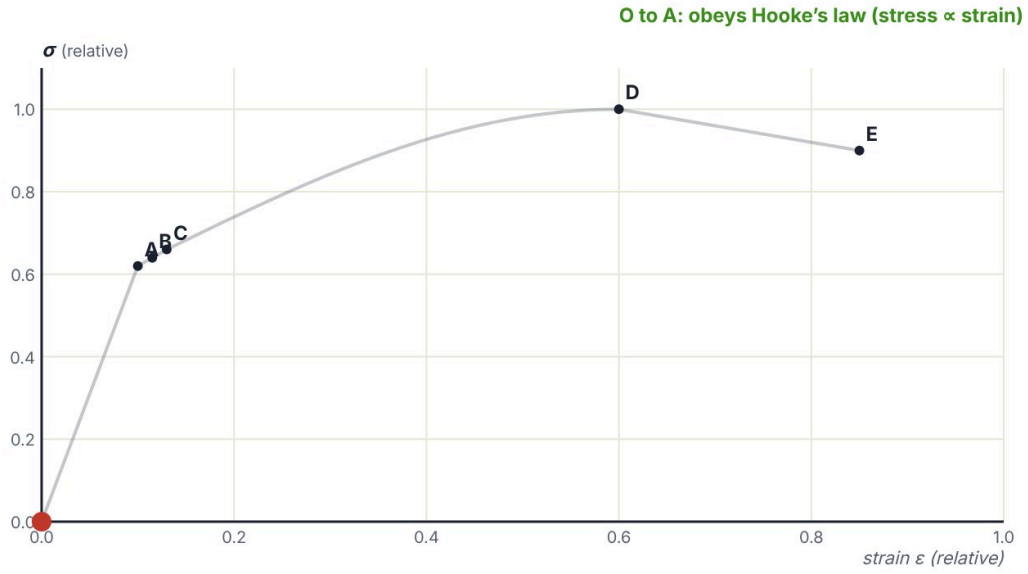
Polymer: chains uncoil

- **Ductile metals**: atoms are held by a "sea" of shared electrons that bond equally in every direction. Under tension whole planes of atoms slide past one another (dislocation motion) without breaking the metallic bond, so the metal stretches a long way before fracture.
- **Brittle materials**: strong ionic or covalent bonds hold atoms rigidly and do not allow planes to slide. The bonds stretch only slightly; when those along one plane reach their limit they all snap together and a crack runs straight through.
- **Polymers**: long tangled chain molecules held together by weak van der Waals forces. A small force uncoils and straightens the chains (large extension for little force). Once the chains are aligned, the force acts on the strong covalent bonds along the backbone, so the graph becomes steep: the S-shaped curve.

Simulation

Stretch a wire: stress–strain explorer

Choose a material and slowly increase the load. The point moves along the curve and the region is named. Release the load after passing the elastic limit to see the permanent extension.



material	stress now	strain now	permanent strain
Ductile metal (copper)	**0.00**	**0.000**	**0.000**

This is a still picture. Open PhysiQ › Notes › Semester 1 › Chapter 9 to use the simulation.

Young's modulus

Young's modulus, Y (or E)

is the **ratio of tensile stress to tensile strain**, provided the limit of proportionality is not exceeded.

$$Y = \frac{\sigma}{\varepsilon} = \frac{F/A}{e/l_0} = \frac{Fl_0}{Ae}$$

dimension $\mathrm{M\,L^{-1}\,T^{-2}}$; unit $\mathrm{N\,m^{-2}}$ = Pa

- Young's modulus describes the elastic properties of a material.
- The larger Y, the harder it is to stretch the material.

Material	Y / 10^{10} Pa
Aluminium	6.9
Copper	11.7
Steel	20
Nylon	0.37
Glass	7.0

Young's modulus and Hooke's law for a wire

Young's modulus: $Y = \frac{Fl_0}{Ae} \Rightarrow F = (\frac{YA}{l_0})e$ (i)

Hooke's law: $F = ke$ (ii)

Comparing (i) and (ii):

$$k = \frac{YA}{l_0}$$

force constant (spring constant) of the wire

9.3 Strain energy

Strain energy

is the **elastic potential energy stored in a stretched (or compressed) material**: the work done on the wire in stretching it.

Up to the limit of proportionality, F increases linearly from 0 to F as the extension goes from 0 to e.

Work done = area under the F–e graph = $\frac{1}{2}Fe$, or $W = \int_0^e kx\,\mathrm{d}x = \frac{1}{2}ke^2$

$$U = \tfrac{1}{2}Fe = \tfrac{1}{2}ke^2$$

strain energy = area under the force–extension graph

Divide by the volume Al_0: $\frac{U}{Al_0} = \frac{1}{2}\,\frac{F}{A}\,\frac{e}{l_0} = \frac{1}{2}\sigma\varepsilon$

Strain energy per unit volume = $\frac{1}{2}\sigma\varepsilon$ = area under the stress–strain graph

Example 9.1

A wire of length 1.0 m stretches by 0.6 mm when a 20 kg mass hangs on it. The radius of the wire is 1 mm. Calculate (a) the stress, (b) the strain, (c) the Young's modulus, (d) the force constant of the wire. (g = 9.81 m s^{-2})

Solution

l_0 = 1.0 m, e = 0.6 × 10^{-3} m, m = 20 kg, r = 1 × 10^{-3} m

(a) $\sigma = \dfrac{mg}{\pi r^2} = \dfrac{(20)(9.81)}{\pi(1 \times 10^{-3})^2} = \mathbf{6.24 \times 10^7\ Pa}$

(b) $\varepsilon = \dfrac{e}{l_0} = \mathbf{6.0 \times 10^{-4}}$

(c) $Y = \dfrac{\sigma}{\varepsilon} = \mathbf{1.04 \times 10^{11}\ Pa}$

(d) $k = \dfrac{Y(\pi r^2)}{l_0} = \mathbf{3.27 \times 10^5\ N\ m^{-1}}$

Example 9.2

A bridge support cable has cross-sectional area 0.0085 m^2 and length 35 m. It is made of high-tensile steel with Young's modulus 2.8 × 10^{11} Pa. The tension in the cable is 720 kN. Calculate (a) the extension, (b) the strain energy stored.

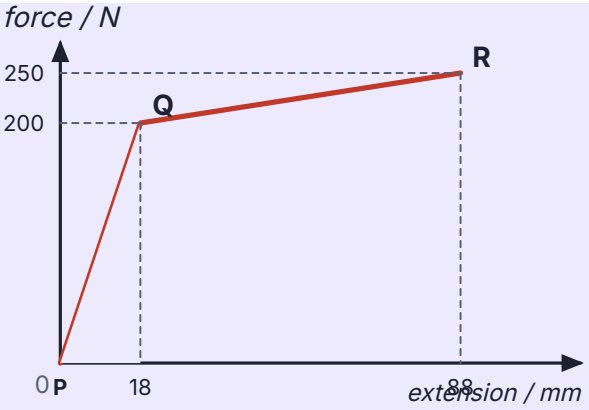
Solution

(a) $Y = \dfrac{Tl_0}{Ae} \Rightarrow e = \dfrac{Tl_0}{YA} = \dfrac{(720 \times 10^3)(35)}{(2.8 \times 10^{11})(0.0085)} = \mathbf{0.0106\ m}$

(b) $U = \frac{1}{2}Te = \frac{1}{2}(720 \times 10^3)(0.0106) = \mathbf{3.82\ kJ}$

Exercises: deformation of solids

1. A wire of unstretched length 2.00 m is stretched by a tensile force. Its force–extension graph has two straight sections: PQ from 0 to 200 N at 18 mm, and QR from 200 N to 250 N at 88 mm. Calculate (a) the strain in the wire at Q and at R, (b) the strain energy stored in the wire at Q, (c) the work done in stretching the wire from P to R.



Answer

(a) 0.009 at Q; 0.044 at R (b) ½(200)(0.018) = 1.8 J (c) 1.8 + ½(200 + 250)(0.070) = 17.55 J

The slide asks for "strain energy" in (a); the answers 0.009 and 0.044 are the strains.

FULL SOLUTION

L = 2.00 m. From the graph: Q is at (18 mm, 200 N) and R is at (88 mm, 250 N).

(a) Strain = e/L

At Q: 0.018 / 2.00 = **0.009**

At R: 0.088 / 2.00 = **0.044**

(b) Strain energy = area under the graph from P to Q (a triangle): ½Fe = ½(200)(0.018)

= **1.8 J**

(c) Work from P to R = area P to Q + area Q to R (a trapezium)

= 1.8 + ½(200 + 250)(0.088 − 0.018) = 1.8 + 15.75

= **17.6 J** (17.55 J)

2. A metal wire 75.0 cm long and 0.13 cm in diameter stretches 0.0350 cm when an 8.00 kg load hangs on it. Find the stress, the strain and the Young's modulus.

Answer

5.91×10^{7} N m^{-2}; 4.67×10^{-4}; 1.27×10^{11} N m^{-2}

FULL SOLUTION

$A = \pi d^2/4 = \pi(0.13 \times 10^{-2})^2/4 = 1.327 \times 10^{-6}$ m^2; F = mg = 8.00 × 9.81 = 78.5 N

Stress $\sigma = F/A = 78.5 / (1.327 \times 10^{-6})$

= **5.91×10^{7} N m^{-2}**

Strain $\varepsilon = e/L = 0.0350 / 75.0$

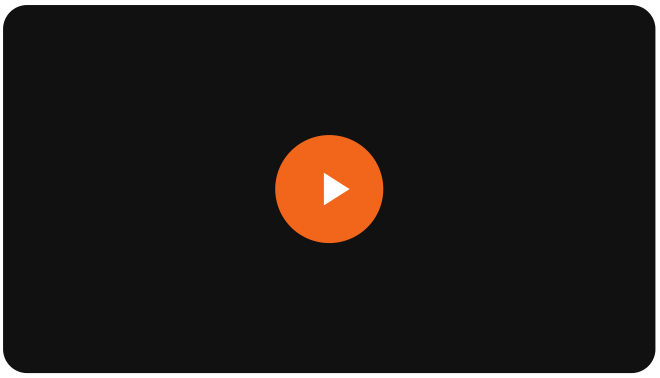
= **4.67×10^{-4}**

Young's modulus $E = \sigma/\varepsilon = 5.91 \times 10^{7} / 4.67 \times 10^{-4}$

= **1.27×10^{11} N m^{-2}**

Physics in Real Life

Real-world videos for this chapter. Tap a picture to play it here.



LABORATORY DEMONSTRATION

Tensile test: stress-strain diagram · UPV

Open in YouTube ↗

A real tensile test

A steel specimen is stretched to breaking: identify the elastic region, the yield point and plastic (permanent) deformation on the stress–strain graph.